

MANIPULATE VIEW

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The Big Picture



Manipulate

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→ Item Reduction

→ Zoom

Geometric or Semantic



→ Pan/Translate



→ Constrained



→ Attribute Reduction

→ Slice



→ Cut



→ Project





Why Change?

Why Change



- Datasets are often sufficiently large and complex that showing everything at once in a single static view would lead to overwhelming visual clutter
- Changing the view over time is the most obvious, most popular, and most flexible choice in vis design
- The most fundamental breakthrough of vis on a computer display compared with printed on paper is the possibility of interactivity



Change View over Time

Introduction



The possibilities for how the view changes can be based on any of the other design choices of how to construct an idiom:

- change the encoding
- change the arrangement
- change the order
- change the viewpoint
- change which attributes are filtered
- change the aggregation level, and so on



Select Elements

Selection Design
Choices

Highlighting

Selection Outcomes

Navigate:
Changing
Viewpoint

Geometric Zooming

Semantic Zooming

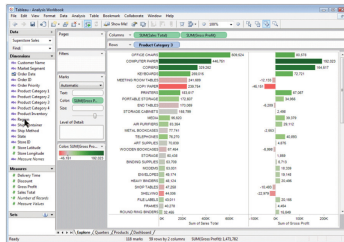
Constrained Navigation

Navigate:
Reducing
Attributes

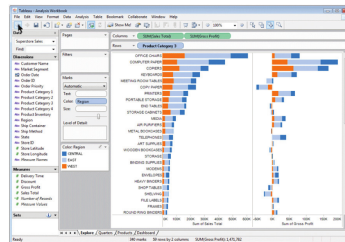
Slide

Cut

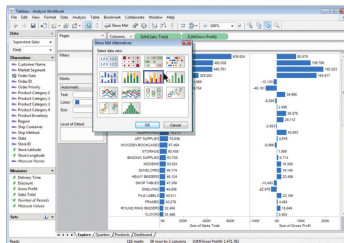
Project



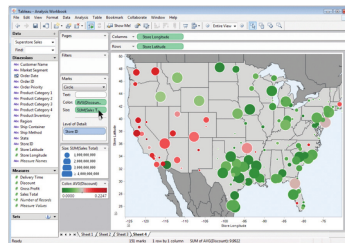
(a)



(b)



(c)

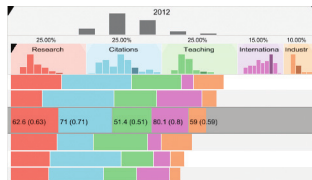


(d)

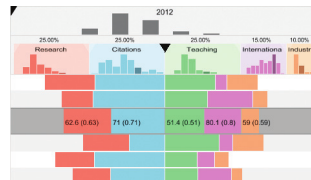


Example: LineUp (cont.)

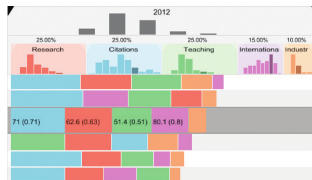
- Changing the alignment in Lineup. (a) Classical stacked bars. (b) Diverging stacked bars. (c) Ordered stacked bars. (d) Separately aligned bars: small multiple bar charts.



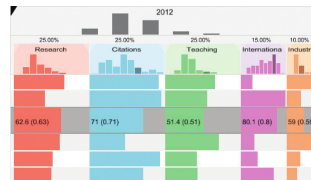
(a)



(b)



(c)



(d)

Example: LineUp (cont.)

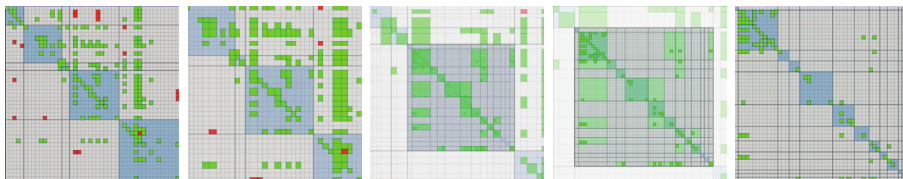


System	LineUp
What: Data What: Derived	Table. Ordered attribute: weighted combination of selected attributes.
How: Encode How: Manipulate Why: Task	Stacked bar charts, slope graphs. Reorder, realign, animated transitions. Compare rankings, distributions.

Example: Animated Transitions



- One of the best-justified uses of animation is the idiom of **animated transition**, where a series of frames is generated to smoothly transition from one state to another
- Frames from an animated transition showing zoom between levels in a compound network arranged as an adjacency matrix



Example: Animated Transitions (cont.)



Idiom	Animated Transitions
What: Data How: Manipulate	Compound network. Change with animated transition. Navigate between aggregation levels.



Select Elements

- Selection Design Choices
- Highlighting
- Selection Outcomes

Introduction



- Allowing users to select one or more elements of interest in a vis is a fundamental action that supports nearly every interactive idiom.
- The output of a selection operation is often the input to a subsequent operation. In particular, the change choice is usually dependent on a previous select result



Selection Design Choices

A fundamental design choice with selection is what kinds of elements can be selection targets

- The most common element is **data items**, and another element is links in the case of network data.
- It's also common to be able to select by **levels** within an attribute.

The number of independent selection types is also a design choice

- In the simplest case, there is only one kind of selection: elements are either **selected** or **not selected**
- It's also common to support two kinds of selection: a **mouse click**, a **mouse hover**

Selection Design Choices (cont.)



Another design choice is how many elements can be in the selection set

- Should there be exactly one?
- Can there be many?
- Is a set of zero elements acceptable?
- Is there a primary versus secondary selection?

Highlighting



- The action of selection is very closely tied to **highlighting**, where the elements that were chosen are indicated by changing their visual appearance in some way
- Selection should trigger highlighting in order to provide users with immediate visual feedback, to confirm that the results of their operations match up with their intentions

Design Choices

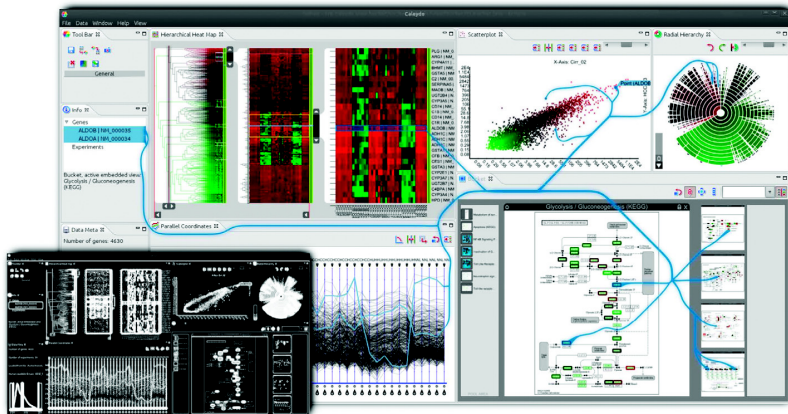


- Two different design choices can be independently varied:
 1. The interaction idiom of how the user selects elements
 2. The visual encoding idiom of how the selected elements are highlighted.
- For data items, nearly any change of visual encoding strategy can be used for highlighting
 1. Changing their color
 2. Highlight with an outline
 3. Use the size channel
 4. Use motion coding
 5. Another different design choice for highlighting is to add connection marks between the objects in the selection set

Example: Context-Preserving Visual Links



- The context-preserving visual links idiom is an example of the design choice to coordinate between views by explicitly drawing links as connection marks between items and regions of interest



Example: Context-Preserving Visual Links (cont.)



Idiom	Context-Preserving Visual Links
What: Data How: Encode How: Manipulate (How: Coordinate)	Any data. Any encoding. Highlight with link marks connecting items across views. Select any element. Juxtaposed multiple views.



Selection Outcomes

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Attributes

Slice

Cut

Project

- The outcome of selection is often additional actions, beyond simply highlighting.
- Selection is often a first step in a multistage sequence, allowing the user to indicate specific elements that should be the target of the next action
 - Selected items might be filtered or aggregated, or their visual encoding could be changed
- A selected set of regions or items might be reordered within a view



Navigate: Changing Viewpoint

- Geometric Zooming
- Semantic Zooming
- Constrained Navigation

Introduction



- Large and complex datasets often cannot be understood from only a single point of view.
- The spatial layout is fixed and navigation acts as a change of the **viewpoint**
- When a **camera viewpoint** is changed, the set of items visible in the **camera frame** also changes
- The term **navigation** refers to changing the point of view from which things are drawn. Navigation can be broken down into three components
 - The action of **zooming** moves the camera closer to or farther from the plane.
 - The action of **panning** the camera moves it parallel to the plane of the image.
 - The action of **rotating** spins the camera around its own axis.

Geometric Zooming



- An intuitive form of navigation is **geometric zooming** because it corresponds almost exactly with our real-world experience of walking closer to objects, or grasping objects with our hands and moving them closer to our eyes

Semantic Zooming

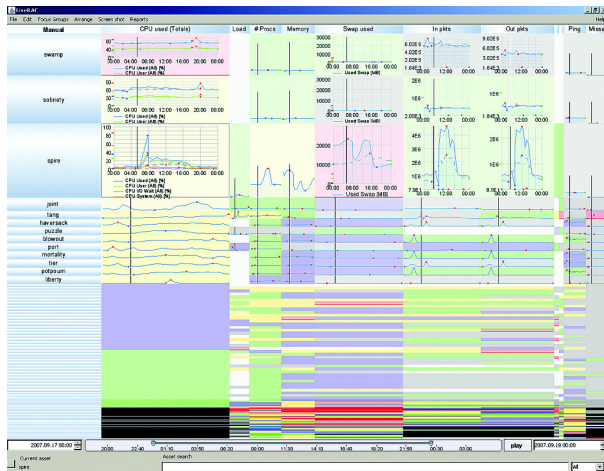


- With **semantic zooming**, the representation of the object adapts to the number of pixels available in the image-space region occupied by the object
- The visual appearance of an object can change subtly or dramatically at different scales.



Example: LiveRAC

- LiveRAC uses semantic zooming within a stretchable grid of timeseries line charts



Constrained Navigation



- With **unconstrained navigation**, the camera can move anywhere at all.
- Navigating with the six degrees of freedom available in 3D space is not a skill that most people have acquired.
- **Constrained navigation** idioms have some kind of limit on the possible motion of the camera.



Navigate: Reducing Attributes

- Slice
- Cut
- Project

Introduction

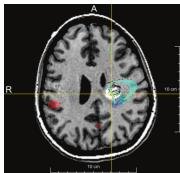


- The geometric intuitions that underlie the metaphor of navigation with a virtual camera also lead to a set of design choices for reducing the number of attributes: slice, cut, and project
- Although these idioms are inspired by ways to manipulate a virtual camera in a 3D scene, they also generalize to higher-dimensional spaces

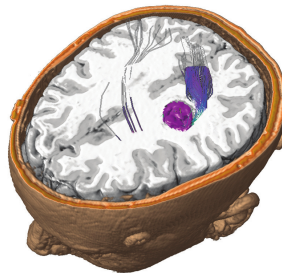
Slice



- With the **slice** design choice, a single value is chosen from the dimension to eliminate, and only the items matching that value for the dimension are extracted to include in the lower-dimensional slice.
- The slice choice eliminates a dimension/attribute by extracting only the items with a chosen value in that dimension. The cut choice eliminates all data on one side of a spatial cutting plane. (a) Axis-aligned slice. (b) Axis-aligned cut



(a)



(b)



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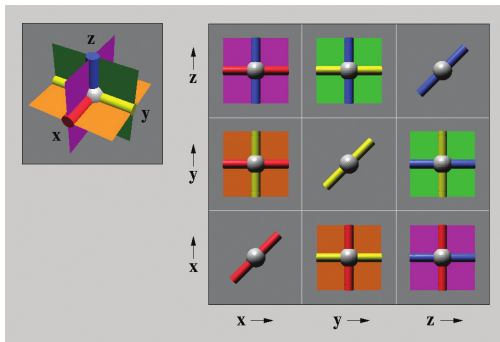
Slice

Cut

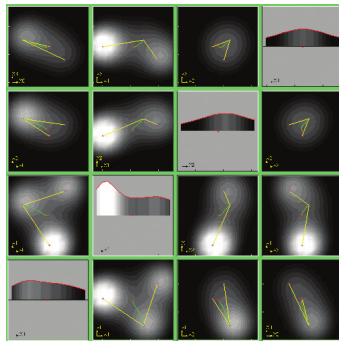
Project

Example: HyperSlice

- The HyperSlice system uses the design choice of slicing for attribute reduction to display abstract data with many attributes: scalar functions with many variables



(a)



(b)

Cut



- The interaction idiom of **cut** allows the user to position a plane that divides the viewing volume in two, and everything on the side of the plane nearest to the camera viewpoint is not shown
- This **cutting plane** can either be axis aligned or arbitrarily oriented

Project



With the **project** design choice, all items are shown, but without the information for specific dimensions chosen to exclude

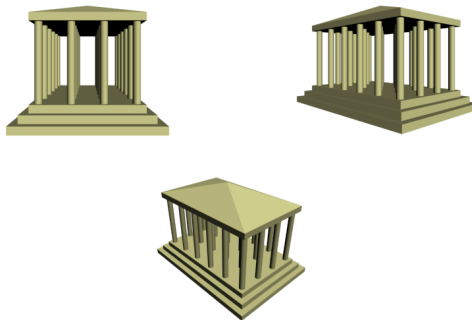
- **Orthographic projection**





Project (cont.)

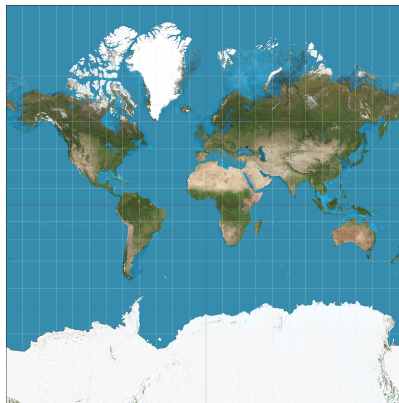
- **Perspective projection:** while a great deal of information about depth is lost, some is retained through the **perspective distortion** foreshortening effect, where distant objects are closer together on the image plane than nearby objects would be



Project (cont.)



- Many map projections from the surface of the Earth to 2D maps have been proposed by cartographers, including the well-known **Mercator projection**



References



Goodfellow, I., Bengio, Y., and Courville, A. (2016).

Deep learning.

MIT press.



Munzner, T. (2014).

Visualization analysis and design.

CRC press.



Russell, S. and Norvig, P. (2016).

Artificial intelligence: a modern approach.

Pearson Education Limited.



Ward, M. O., Grinstein, G., and Keim, D. (2015).

Interactive data visualization: foundations, techniques, and applications.

CRC Press.